Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 26: There Is One

Existence Proofs

Chartrand & Zhang, Section 3.6, pp. 105–108

Name: _____ Date: _____

Where we are

Every method so far proves a statement beginning $\forall$. Today's statements begin $\exists$ — and they are usually the easiest to prove, because one witness does the whole job. They are also the source of one of the strangest proofs in the course.

By the end of today you should be able to

1. Prove an existence statement by exhibiting a witness.
2. Disprove an existence statement.
3. Explain how a proof can establish existence without naming the object.

1. One witness proves it

Existence proof

To prove $\exists x \in S,\ R(x)$, produce one **specific** element $a \in S$ and verify that $R(a)$ is true.

Result 3.34. There exists an integer n with $2 - n^2 > 0$.

Proof. Let $n = 1$. Then $2 - 1^2 = 1 > 0$. ■

Result 3.36. There exists a real number x with $x^2 - 8 = 0$.

Proof. Let $x = \sqrt{8}$. Then $(\sqrt{8})^2 - 8 = 0$. ■

Verify, do not just announce

"Let $n = 1$" is half a proof. The other half is *showing* $R(1)$ holds. Writing the witness without the check is the most common way existence proofs lose points.

2. Disproving existence is the hard direction

To show $\exists x \in S,\ R(x)$ is false, you must show $\forall x \in S,\ \sim R(x)$ — an argument about every element. No amount of testing does it.

Example 3.39. Disprove: *there exists* $x \in \mathbb{R}$ *with* $x^4 + 2 = 2x^2$.

Solution. Let $x \in \mathbb{R}$. Rewrite and complete the square:

$$x^4 - 2x^2 + 2 = \left(x^4 - 2x^2 + 1\right) + 1 = \left(x^2 - 1\right)^2 + 1 \ \geq\ 0 + 1 = 1 > 0.$$

So $x^4 - 2x^2 + 2 \neq 0$ for every real x, and the statement is false.

This is Week 4's asymmetry once more: one example proves $\exists$; refuting it takes an argument covering the whole domain.

3. Mixed quantifiers

Statement	Proof
For every irrational s, there is a rational r with rs irrational. (Result 3.40)	Given s, take $r = 1$; then $rs = s$ is irrational.
There is a rational r such that for every irrational s, rs is rational. (Result 3.41)	Take $r = 0$; then $rs = 0$ for every s.

Why the order matters in the proof

In the first, r is chosen *after* s, so it may depend on s — though here it happens not to. In the second, r must be chosen *first* and work for every s at once, which is why only 0 can do it.
The witness has to be chosen at the point the quantifier appears. That is Week 4, now as a proof technique.

4. A proof that does not tell you the answer

Result 3.38

There exist irrational numbers a and b such that a^b is rational.

Proof. Consider the number $\sqrt{2}^{\sqrt{2}}$. It is either rational or irrational.

Case 1: $\sqrt{2}^{\sqrt{2}}$ is rational. Take $a = b = \sqrt{2}$, both irrational, and we are done.

Case 2: $\sqrt{2}^{\sqrt{2}}$ is irrational. Take $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$, both irrational. Then

$$a^b = \left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\cdot\sqrt{2}} = \sqrt{2}^{\,2} = 2,$$

which is rational. Either way, suitable a and b exist. ■

Read that again

The proof never says *which* case holds. We have proved that a pair exists without being able to name it — one of two candidates works, and we do not know which.
This is a **non-constructive** existence proof, and it is built from last Friday's proof by cases. (For the record, $\sqrt{2}^{\sqrt{2}}$ is irrational, so Case 2 is the true one — but proving that is far beyond this course.)

Constructive versus non-constructive

A **constructive** proof hands you the object: "let $n = 1$." A **non-constructive** proof shows it must exist without producing it.
In computing, the difference is enormous: a constructive proof is an algorithm, and a non-constructive one is not.

Exit ticket

1. Prove: there exists an integer n with $n^2 - 5n + 6 = 0$.

2. What kind of argument is needed to *disprove* a statement beginning $\exists$?

Before Friday

Read Section 3.7, "Proof by Contradiction," pp. 108–111.

Self-check (not collected). Section 3.6, Exercises 3, 5, 7, and 9 — odd-numbered, answers in the back.

Homework 9 is due Friday, October 23, 11:59 p.m.

Questions I still have